

Risk Radar

Reads every project update and surfaces the risks, blockers and dependencies that threaten delivery — in plain English, with the reasoning shown.

Challenge attempted: #1 — Risk Radar. It also delivers #3 — Status Summarizer — as a built-in output of the same workflow.

Live prototype <https://careemdemo-scnnzwjvpfmggqug6us34k.streamlit.app/>

Source code <https://github.com/Demonmonk/CareemDemo>

Public dataset https://github.com/Demonmonk/CareemDemo/blob/claude/keen-carson-Xbmqu/data/project_updates.csv

The live app runs instantly in any browser — no login, no setup.

100-word summary

Risk Radar is an AI early-warning system for delivery. It ingests the messy status updates a program manager drowns in — standups, emails, Jira comments — and uses the **Claude API** to classify each into a structured signal: blocker, risk, dependency or on-track, with a severity, the reason it was flagged, and a recommended next step. Signals roll up into a RAG health per project and a director-ready status digest. Built with structured outputs, prompt caching and a rule-based fallback (so it runs with no key), plus a hard spend cap for safe public use. Accuracy is measured against planted ground-truth labels.

The program-manager lens

Delivery doesn't fail in one big bang — it slips one buried update at a time. By the time it reaches a steering deck, the milestone is gone. Risk Radar is the early-warning layer a PMO actually needs:

- **Surfaces risk early** — flags trouble the day it's written, not at sprint's end.
- **Makes dependencies and blockers explicit** — the things that quietly sink cross-team delivery.
- **Speaks the language of status** — RAG health, owners and concrete next steps; the digest is the weekly update, written for you.
- **Shows its reasoning** — every flag comes with *why*, so it earns trust instead of being a black box.

Under the hood

- **Claude API** for classification and summarization, via **structured outputs** — reliable typed JSON, not free text to parse.
- **Prompt caching** on the static instructions, and a deliberately fast, low-cost model for a high-volume, well-scoped task.
- **Graceful rule-based fallback** — fully functional with no API key, so anyone can evaluate it instantly.
- **Production guardrails** — server-side-only secrets, a hard shared spend cap, and result caching so nothing is re-charged.
- **Evaluated, not asserted** — hidden ground-truth labels the model never sees turn accuracy into a measured number (~94%).

Portfolio snapshot — live output

Generated from the prototype's classification of the dataset. Projects are ranked worst-first.

Project	Program	Health	Blk	Rsk	Dep
PCI-DSS Compliance (Careem Pay)	Trust & Safety	● RED	3	2	1
Send & Request Money	Careem Pay	● RED	2	2	2
Wallet Top-up & Cards	Careem Pay	● RED	1	3	1
15-min Grocery Fulfilment	Quik	● RED	1	3	1
Restaurant Onboarding Portal	Careem Food	● AMBER	0	3	0
Super-App Kubernetes Migration	Platform & Infrastructure	● GREEN	0	1	1
Captain Earnings & Payouts	Captain Experience	● GREEN	0	0	1
Live Order Tracking	Careem Food	● GREEN	0	0	0
Careem Plus Revamp	Growth & Subscriptions	● GREEN	0	0	0
API Gateway Consolidation	Platform & Infrastructure	● GREEN	0	0	0
Captain App 4.0	Rides	● GREEN	0	0	1
Surge Pricing Engine v2	Rides	● GREEN	0	0	0

Example — one raw update, read by the AI

“Critical blocker on the earnings dashboard — a data-migration step is corrupting records and we’ve had to halt the rollout entirely.”

● **BLOCKER · CRITICAL**

Why flagged: The wording “blocker” means work is stopped until someone acts.

Suggested action: Escalate to the PMO and identify the owner to unblock.

30-second evaluation guide

- **Overview** — projects ranked worst-first; open a red one to see *why* each issue was flagged.
- **Project view** — each raw update shown beside the AI's read, plus a trend line and a status digest.
- **How it works** — the workflow, and the AI's measured accuracy vs. the hidden ground-truth labels.

Dataset: synthetic, illustrative mock data hand-generated for this demo, themed around Careem-style lines of business (Careem Pay, Rides, Careem Food, Quik, platform, compliance). Not real Careem data and not affiliated with, endorsed by, or sourced from Careem. No confidential information.